

# Jason Hong

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## EDUCATION

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### Massachusetts Institute of Technology

Cambridge, MA

*Bachelor of Science in Computer Science and Engineering*

*Sept 2025 – May 2028*

- GPA: 5.0/5.0
- Transferred from community college; first-generation college student
- Coursework: Introduction to Algorithms, Intro to CS and Programming (Python), Mathematics for Computer Science

### Fullerton College

Fullerton, CA

*Associate Degree, Computer Science*

*Sept 2024 – May 2025*

- GPA: 4.0/4.0
- Leadership: Math Club President, Physics Club President, Supplemental Instructor

## EXPERIENCE

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### Kyte Dynamics | Aerial Robotics Co.

May 2025 – August 2025

*Software Engineer Intern*

*Los Angeles, CA | Vancouver, BC, Canada*

- Implemented embedded C/C++ PX4 output drivers enabling precise actuator control for rover-drone integration
- Refactored **2,000+ lines** of legacy C/C++ into modular abstractions, improving scalability and maintainability
- Designed CI/CD pipeline custom parameters, **reducing compile time by 50%**
- Built MATLAB/Simulink test benches for IMU sensors to validate performance and prevent failures

### University of California, Los Angeles Geology Research Laboratory

June 2024 – September 2024

*Undergraduate Research Assistant*

*Los Angeles, CA*

- Automated data processing pipelines using Python and Linux shell scripts, **reducing analysis time by 60%**
- Measured and sampled data from mass spectrometer for geological analysis
- Conducted technical training for non-technical staff on Linux systems and command-line tools

## PROJECTS

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### The Circle of Least Confusion | Volunteer Non-profit Organizer

May 2022 – Present

- Led social media campaigns across TikTok, Instagram, and Facebook to **raise and disperse \$35,500+ in grants** for optometry students
- Created and produced **100+ videos, posts, and visual content**, growing audience by **543%** (700 to 4,500+ followers) and increasing engagement by **110%**
- Represented organization at government conferences and student meetings across California to expand program reach

### Autonomous Defense UAV Platform | PX4 Autopilot, RPi 5, Computer Vision

May 2025 – August 2025

- Engineered autonomous UAV system using PX4 autopilot firmware with MAVLink protocol integration
- Developed computer vision pipeline using Intel RealSense D435i and RPi 5, achieving real-time object detection using YOLO models with **30+ FPS processing**

### Unix Shell Implementation | C, Systems Programming

October 2025 – November 2025

- Built a fully functional Unix shell from scratch in C with process management, I/O redirection, and pipeline support
- Implemented job control features including foreground/background processes, signal handling, and built-in commands
- Managed memory allocation and deallocation to prevent leaks, achieving zero memory errors in Valgrind testing

## TECHNICAL SKILLS

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**Languages:** C/C++, Python, MATLAB, Bash

**Frameworks & Libraries:** PyTorch, OpenCV, YOLO, NumPy

**Developer Tools:** Linux, Git, CMake, CI/CD, Neovim